

Handwritten Form Recognition using YOLO and TrOCR with Layout Aware Error Correction

Agenda

- Team Introduction
- Problem Statement
- Methodology
 - Detection Model
 - Field Matching Mechanism
 - OCR Model
 - Error Correction
- Benchmarks
- Final Results
- Challenges and Key Learnings
- Future Work

Team Introduction



Sathishkumar
Sr. Scientist
IGCAR



Anand
Sr. Consultant



Iniya
Data Scientist
Secure Traces



Muthukumar
Scientist, IGCAR

Problem Statement

BIO-DATA			
1	FORM ID	1	
2	CANDIDATE NAME	Dayita Bakshi	
3	FATHER/ HUSBAND NAME	Laksh Bakshi	
4	DATE OF BIRTH	12/27/1975	
5	QUALIFICATION	Post- Graduate	
6	MARITAL STATUS	7	GENDER Female
8	BLOOD GROUP	9	NATIONALITY Indian
10	EXPERIENCE	a. 9 years at Dewan Ltd b. 10 years at Chahal PLC	
11	PRESENT ADDRESS	H.No. 133, Bala Circle, Bongaigaon - 379102	
12	PERMANENT ADDRESS	H.No. 23, Chaudry Path, Nadiad - 559407	
13	CONTACT NUMBER	14	ALTERNATE NO.
	9351045691		9787612803
15	LANGUAGES KNOWN	Marathi, English, Telugu, Hindi	
16	REFERENCES C MOB NO	a. Kritika Brar - 7099406444 b. Tanaki Handa - 8634823848	
17	GOVERNMENT VALID PHOTO IDENTITY (ANY ONE)		
	AADHAAR CARD	358086000000	PAN CARD
18	PLACE	Dehradun	
19	DATE	06/17/2023	



```
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    "Field value": "Dayita Bakshi",
    "Coordinate": [
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      1550,
      485
    ]
  },
  {
    "Field name": "Father/husbandname",
    "Field value": "Laksh Bakshi",
    "Coordinate": [
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      484,
      1617,
      583
    ]
  },
  {
    "Field name": "Dateofbirth",
    "Field value": "12/27/1975",
    "Coordinate": [
      969,
      586,
      1513,
      677
    ]
  }
],
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Related Work

Layout LM

- ✓ Understands semantic structure of the document
- ✓ Proven record on layout recognition tasks
- ✗ Not directly suitable for images with handwritten text
- ✗ Training requires labelled key value pairs

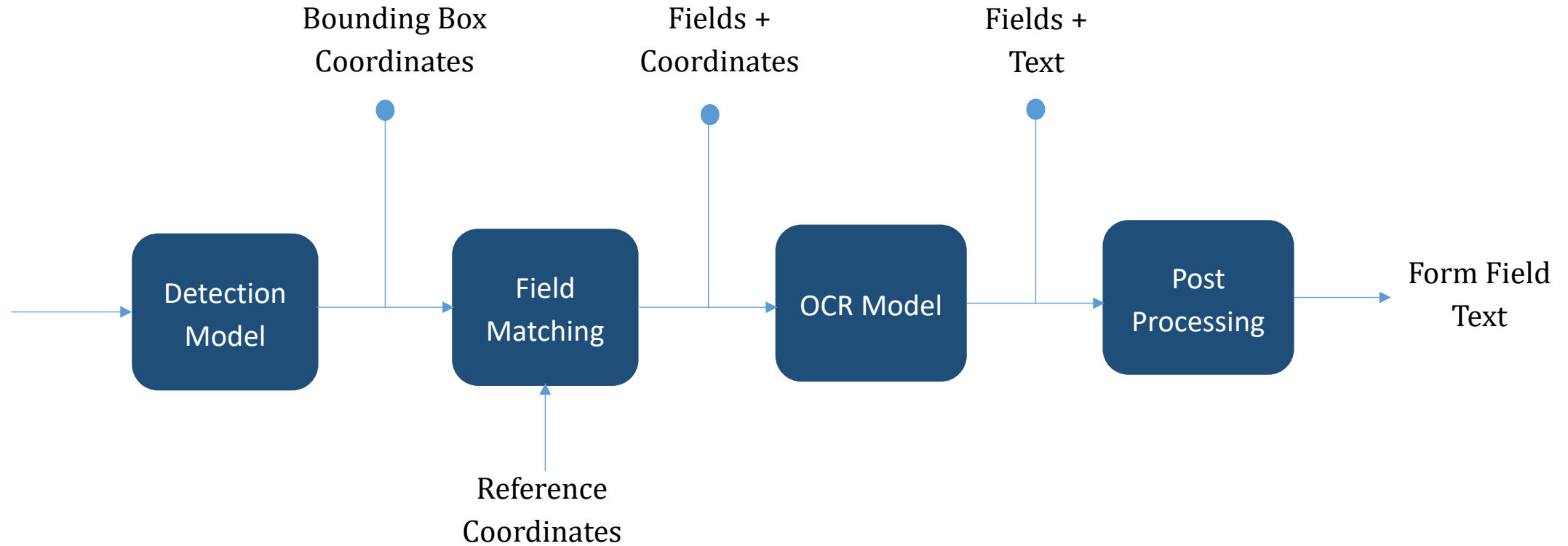
Donut Model

- ✓ Understands semantic structure of the document
- ✓ Proven record in form Recognition tasks
- ✗ Not directly suitable for images with handwritten text
- ✗ Training requires complex hardware

Detection + OCR

- ✓ Simple and Highly modular
- ✓ Performance debugging is possible
- ✗ Requires training of each module individually
- ✗ High time complexity

Model Architecture



Field Detection Model

BIO-DATA

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6	MARITAL STATUS	7	GENDER Female
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10	EXPERIENCE	a. 9 years at Dewan Ltd b. 10 years at Chahal PLC	
11	PRESENT ADDRESS	H.No. 133, Bala Circle, Bongaigaon - 379102	
12	PERMANENT ADDRESS	H.No. 23, Chaudry Path, Nadiad - 559907	
13	CONTACT NUMBER	14	ALTERNATE NO.
	9351045691		9787612803
15	LANGUAGES KNOWN	Marathi, English, Telugu, Hindi	
16	REFERENCES C MOB NO	a. Krihika Brar - 7099406444 b. Tanaki Handa - 8634823848	
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19	DATE	06/17/2023	

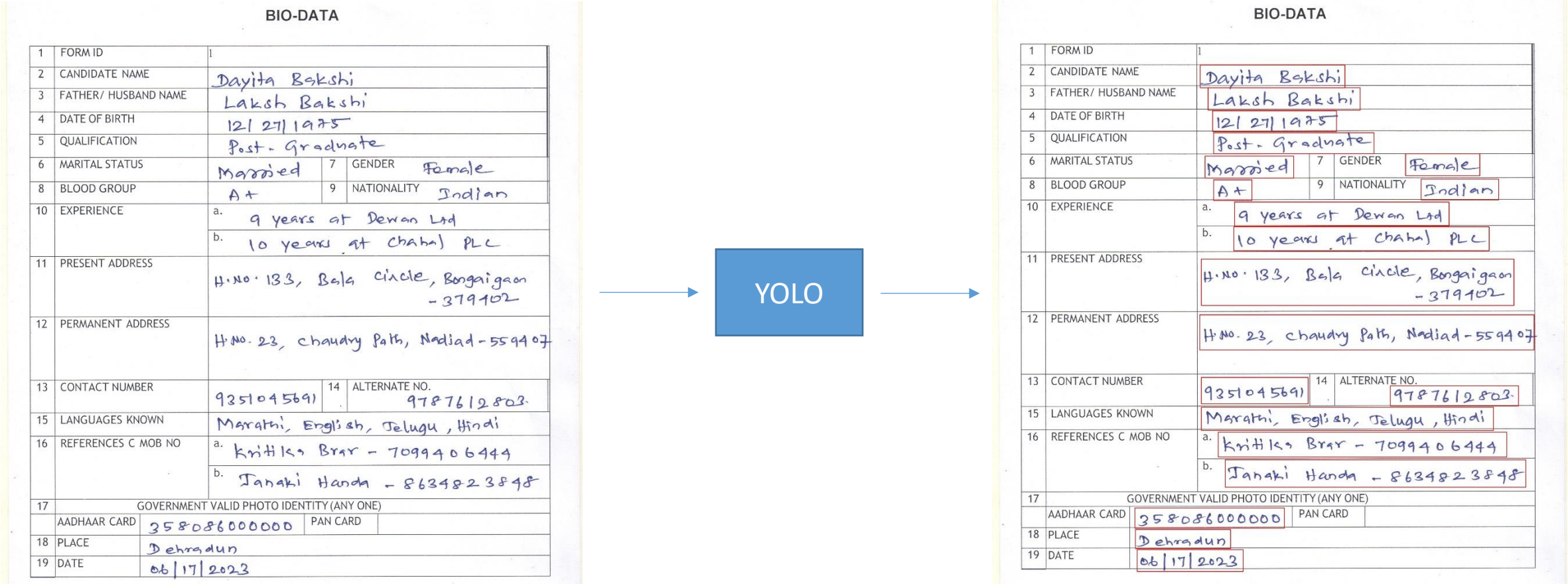
YOLO

Parameters	Value
Version	v8 medium
Parameters	25M
Input Size	640 X 640
Conf	0.25
IOU	0.4

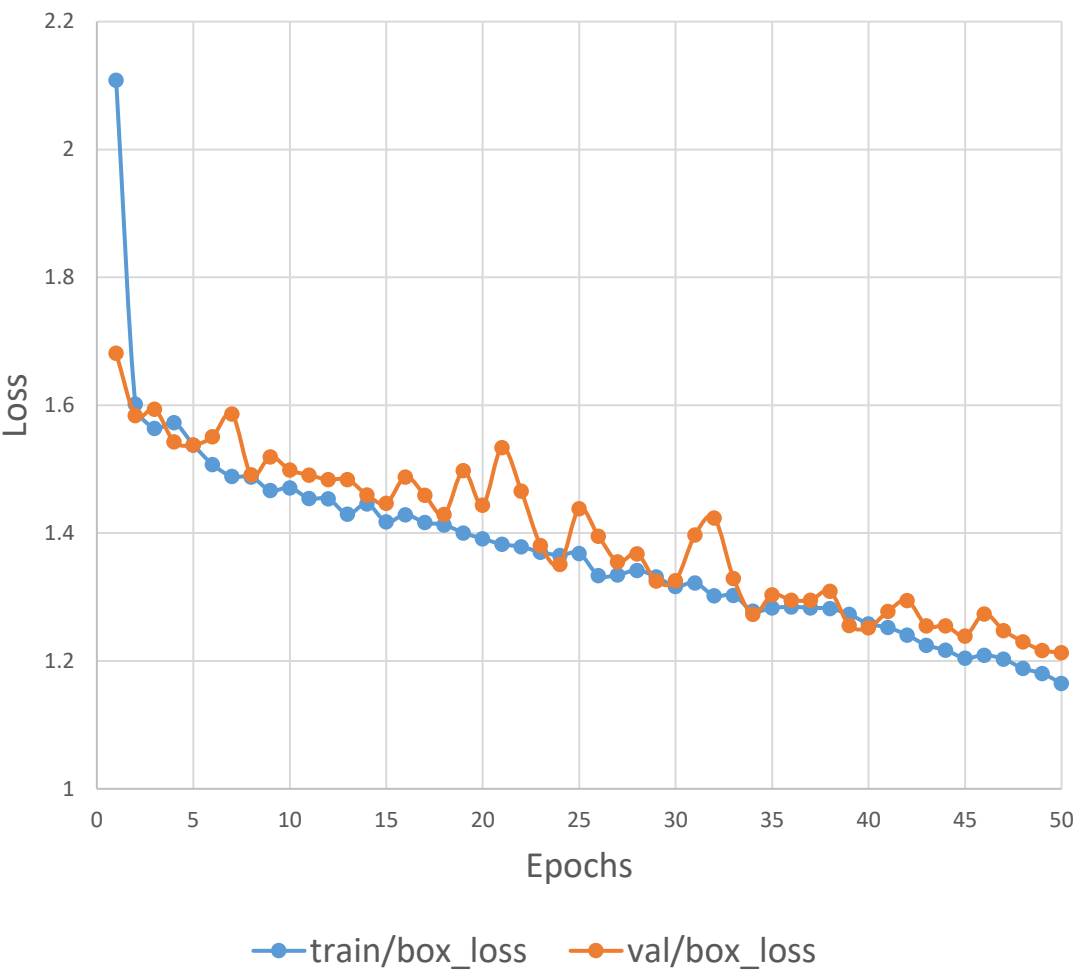
BIO-DATA

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2	CANDIDATE NAME	Dayita Bakshi	
3	FATHER/ HUSBAND NAME	Laksh Bakshi	
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6	MARITAL STATUS	7	GENDER Female
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	AADHAAR CARD	358086000000	PAN CARD
18	PLACE	Dehradun	
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Field Detection Model: YOLO



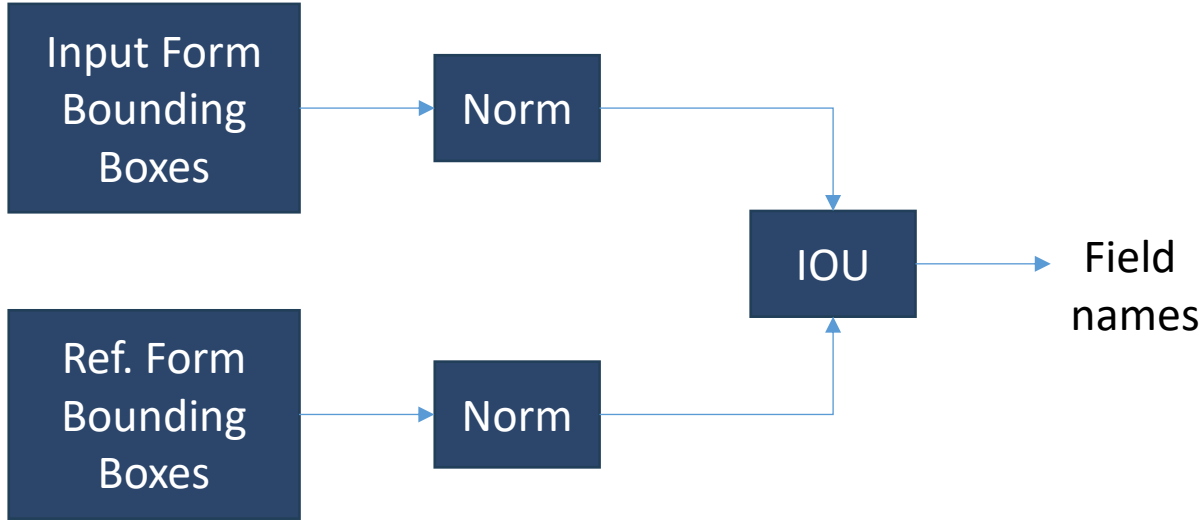
Field Detection Model: YOLO Results



		Actual	
Prediction		Positive	Negative
	Positive	20382	209
	Negative	254	NA

-  mAP@0.5 : 92.6%
-  Precision : 98.9%
-  Recall : 98.7%
-  F1 Score : 98.8%

Field Matching

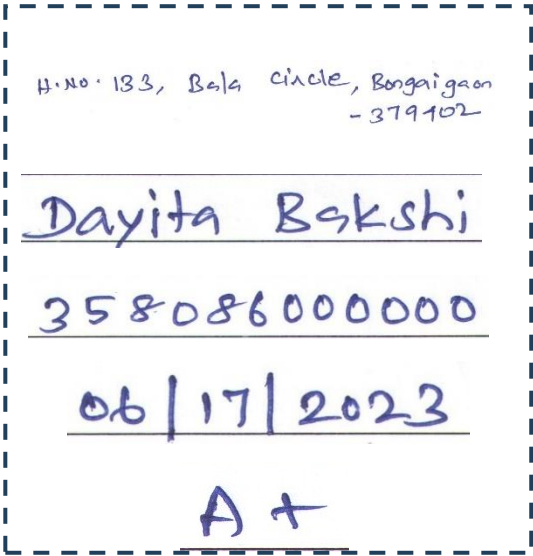


Normalization

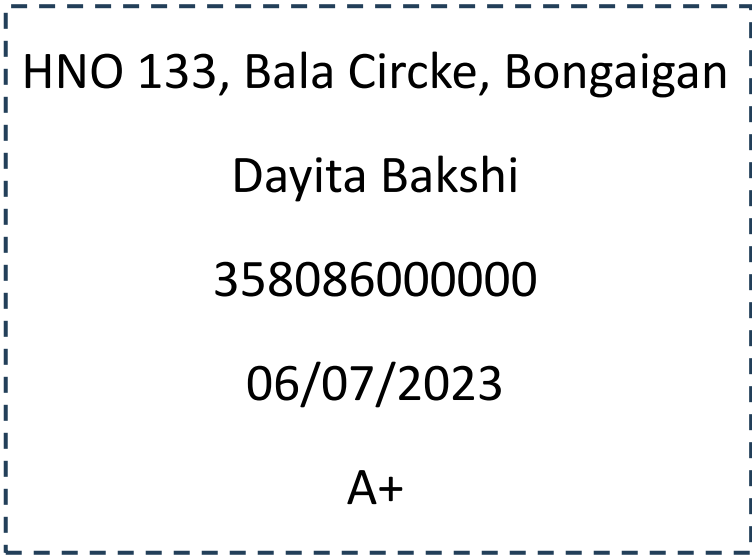
BIO-DATA

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3	FATHER/ HUSBAND NAME	Laksh Bakshi		
4	DATE OF BIRTH	12/27/1975		
5	QUALIFICATION	Post- Graduate		
6	MARITAL STATUS	Married	7	GENDER Female
8	BLOOD GROUP	A+	9	NATIONALITY Indian
10	EXPERIENCE	a. 9 years at Dewan Ltd b. 10 years at Chahal PLC		
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15	LANGUAGES KNOWN	Marathi, English, Telugu, Hindi		
16	REFERENCES C MOB NO	a. Kritika Brar - 7099406444 b. Tanaki Handa - 8634823898		
17	GOVERNMENT VALID PHOTO IDENTITY (ANY ONE)			
	AADHAAR CARD	358086000000	PAN CARD	
18	PLACE	Dehradun		
19	DATE	06/17/2023		

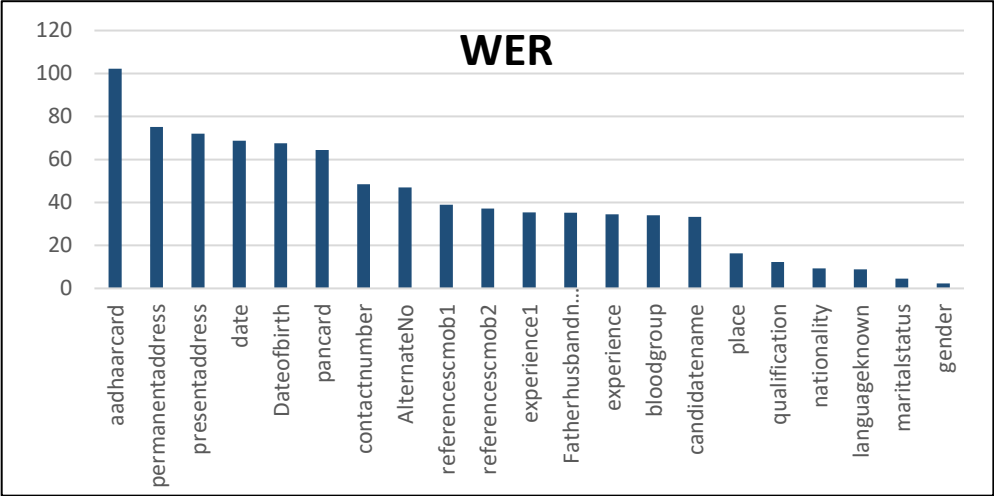
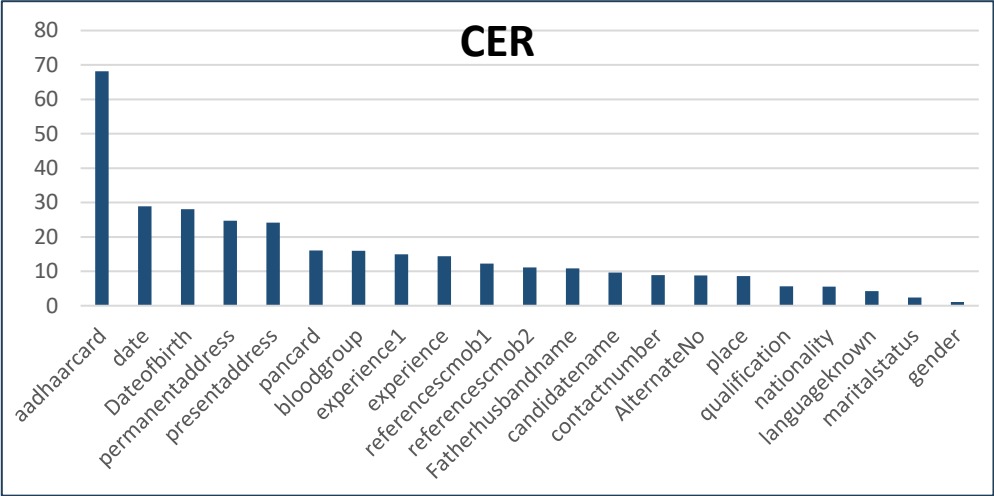
OCR Model



Parameters	Value
Type	ViT
Version	TrOCR-Large
Pre-Trained	IAM Dataset
Parameters	580M
Input Size	1000x1000
IOU	0.4



OCR Results



Layout Aware Error Correction

Fix Incorrect Character

Misrecognized characters are corrected

- In Date, / is detected one
- In alphabet only fields, B/O is detected as 8/0

Limit Field Characters

Limit numerical fields to numbers

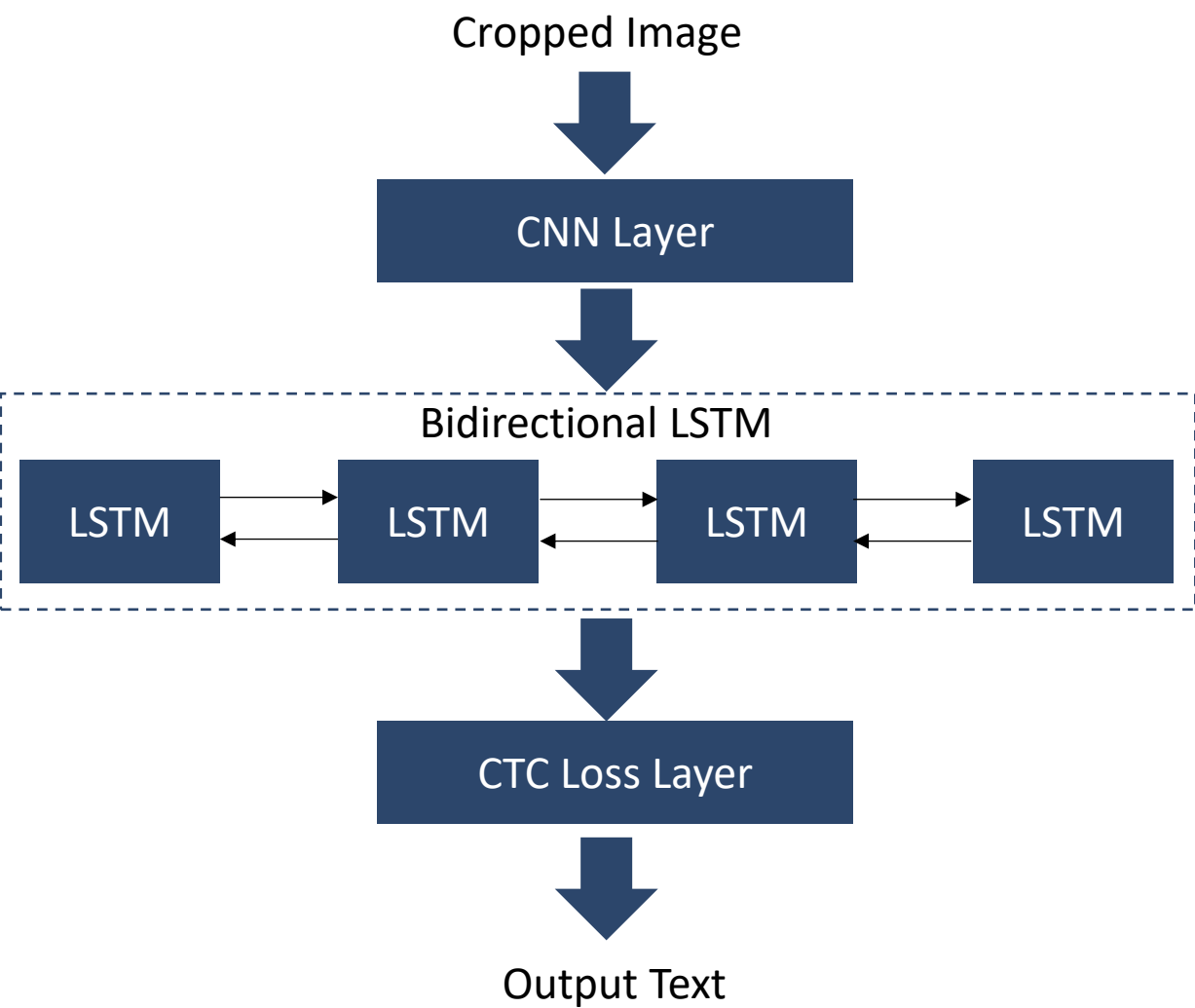
- Aadhar to Numeric only,
- Name to Text only

Autocorrect Text

Fields with predefined values are autocorrected based on training data

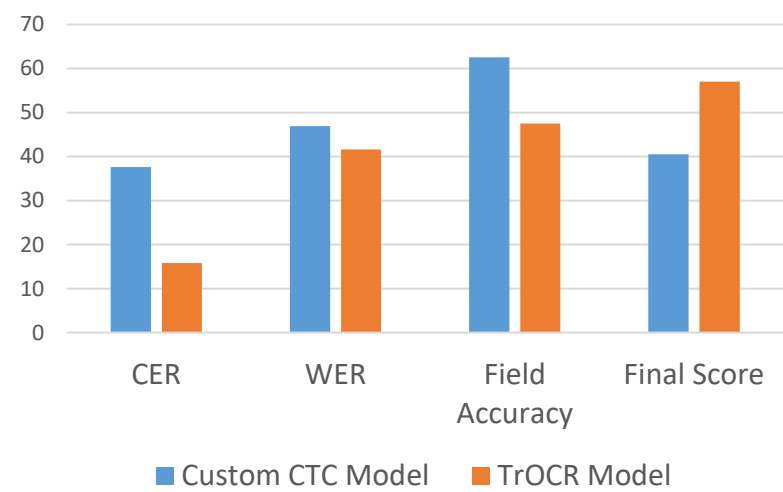
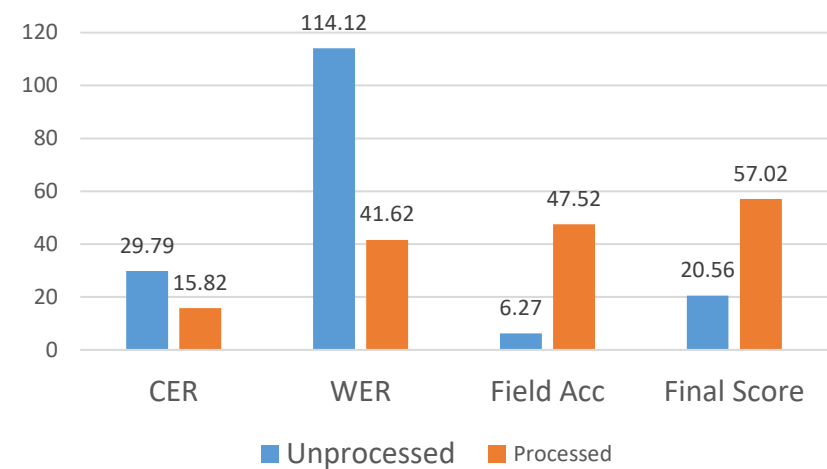
- Nationality: Indian
- Marital Status: Married/Single

Benchmarks: CTC based OCR Model



Parameters	Value
Type	CTC OCR Model
CNN Layers	3
Bi-LSTM Layers	2
Parameters	9M
Batch Size	128
Epochs	250

Benchmark Results



Model	CER	WER	Field Accuracy	Final Score
YOLO + Tesseract	72.54	159.32	2.24	-10.81
YOLO + Custom OCR	37.58	46.93	62.58	40.51
YOLO + TrOCR	29.79	114.12	6.27	20.56
YOLO + TrOCR + Layout Aware Post Processing	15.82	41.62	47.52	57.02

Challenges and Learnings



- Resource constraints in Fine tuning complex models
- Cleaning the dataset
- Variation in the handwriting styles



- Custom built models are flexible though not powerful
- Domain knowledge Matters
- Pre and Post processing improves accuracy

Future Work



- Finetune the architecture of custom OCR Model
- Combine detection and recognition into unified training pipeline
- Cloud deployment
- Extend it to general form recognition

Acknowledgement

- Sincere Thanks to DeHaDo-AI Challenge organizers for the opportunity to work on a real-world handwritten form recognition problem and for curating a valuable dataset.
- Grateful to IIT Jammu for selecting and presenting our work at the NCVPRIPG Conference.